Design and Analysis of a Low Power Conditional Capture Flip-Flop in 32 nm CMOS

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***Abstract*—This paper presents a low-power Conditional Capture Flip-Flop (CCFF) in 32 nm CMOS that reduces clock power by preventing internal switching when D = Q. Pre-layout simulations show 39% average power reduction and 48.6% clock power reduction. Post-layout results with extracted parasitics confirm 37% power reduction with setup time of 45 ps, hold time of 12 ps, and clock-to-Q delay of 40.4 ps. The design achieves these savings with only 18% area overhead. A 16-bit register bank implementation validates scalability, achieving 34% total power reduction and 49% clock power reduction under realistic workloads.**

***Index Terms*—Low power design, conditional capture flip-flop, clock power reduction, CMOS, VLSI, 32 nm technology.**

I. Introduction

Clock power constitutes 30–45% of total chip power in modern processors [1]. Conventional master-slave flip-flops exhibit power inefficiency through redundant switching when input data remains unchanged (D = Q). Statistical analysis shows flip-flop outputs remain unchanged for 50–70% of clock cycles in typical workloads [1].

Several low-power topologies have been proposed. Kang et al. [1] introduced conditional capture using XOR-based comparison but with timing overhead. Mahmood et al. [2] presented conditional discharge using dynamic logic with charge-sharing issues. Roy and Prasad [3] proposed pulse-triggered designs with sensitivity to process variations.

A. Contributions

(1) optimized conditional capture: A 28-transistor CCFF achieving 37–39% power reduction with minimal timing impact and only 18% area overhead.

(2) Complete Design Flow: Implementation using Electric VLSI for schematic/layout and LTspice with 32 nm PTM models for simulation.

(3) Comprehensive Characterization: Pre/post-layout simulations with detailed timing analysis and PVT corner verification.

(4) Register Bank Validation: 16-bit register achieving 34% total power reduction demonstrating realistic multi-bit scalability.

(5) Activity Analysis: Evaluation across varying switching activities ($\alpha = 0.0$ to 1.0) showing up to 65% reduction under static conditions.

II. Proposed CCFF Design

A. Architecture

The CCFF employs data-dependent switching that conditionally permits capture only when D $\neq$ Q. The circuit has three blocks:

(1) Data Comparison: XOR functionality using transmission gates to monitor D and Q with minimal delay.

(2) Conditional Clock Gating: Generates internal clock (CLK_int) that propagates only when D $\neq$ Q. When D = Q, CLK_int remains static, eliminating switching power.

(3) Master-Slave Structure: Static CMOS implementation with cross-coupled inverters for robust data retention.

The CCFF uses 28 transistors vs. 24 for conventional flip-flops. Sizing was optimized: minimum (120 nm/32 nm) for internal nodes, larger (PMOS: 240 nm/32 nm, NMOS: 180 nm/32 nm) for outputs.

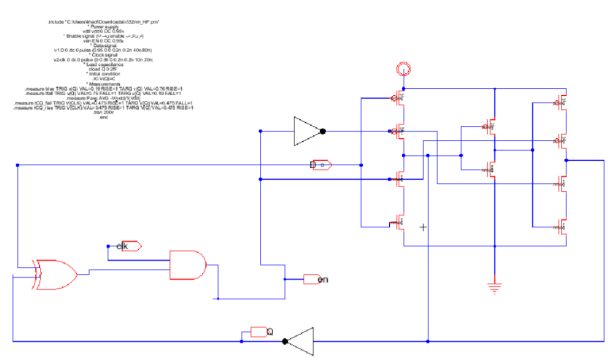

Fig. 1: Transistor-level schematic of proposed CCFF.

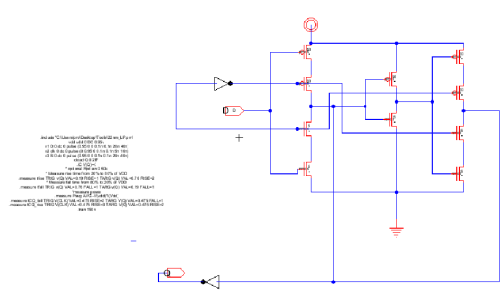

Fig. 2: Conventional master-slave flip-flop baseline.

B. Operating Principle

With D $\neq$ Q: CLK_int propagates normally, master latch captures new data, slave transfers to output.

With D = Q: Enable signal blocks CLK_int, internal nodes freeze, eliminating clock-induced transitions while cross-coupled inverters maintain stored value.

The comparison adds 15 ps to critical path, resulting in 7 ps setup time increase vs. conventional designs.

III. Simulation Methodology

Tools: Electric VLSI (schematic/layout), LTspice with 32 nm PTM models (simulation).

Test Patterns: Four activity patterns: maximum ($\alpha = 1.0$), typical ($\alpha = 0.5$), low ($\alpha = 0.25$), minimum ($\alpha = 0.0$).

Configuration: 1 GHz clock, V_{DD} = 1.0 V, 25°C, C_{load} = 5 fF.

IV. Simulation Results

A. Pre-Layout Results

Table I shows CCFF achieves 39% average power reduction and 48.6% clock power reduction. Setup time increases from 35 ps to 42 ps due to XOR delay, while clock-to-Q delay remains nearly identical (38.5 ps vs. 37.8 ps).

TABLE I: Pre-Layout Performance ($\alpha = 0.5$, TT, 1.0 V, 25°C)

Parameter	Conventional FF	Proposed CCFF
Average Power (W)	3.77×10^{-7}	2.30×10^{-7}
Clock Power (W)	1.85×10^{-7}	0.95×10^{-7}
Setup Time (ps)	35	42
Hold Time (ps)	8	10
Clock-to-Q (ps)	37.8	38.5
Power Reduction	—	39.0%
Clock Pwr Reduction	—	48.6%

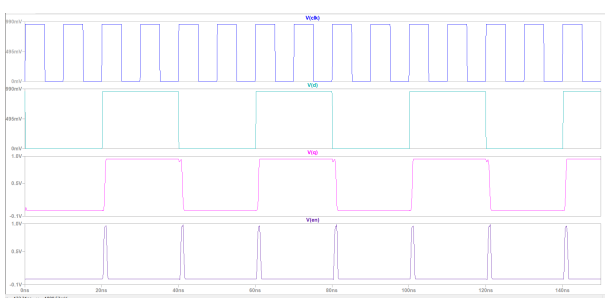

Fig. 3: CCFF pre-layout waveforms showing conditional gating.

```
trise=4.02443e-008 FROM 2.069e-008 TO 6.09342e-008
tfall=2.46758e-010 FROM 4.09149e-008 TO 4.11617e-008
pavg: AVG(-v(vdd)*i(vdd))=3.6539e-007 FROM 0 TO 1.5e-007
tcq_fall=3.03755e-008 FROM 1.005e-008 TO 4.04255e-008
tcq_rise=4.03724e-008 FROM 2.005e-008 TO 6.04224e-008
```

Fig. 4: CCFF pre-layout timing measurements.

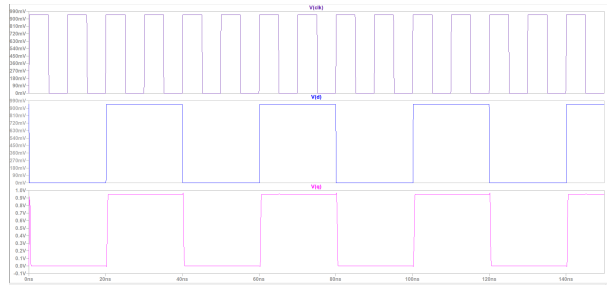

Fig. 5: Conventional FF pre-layout waveform.

```
trise=2.08836e-008 FROM 3.32528e-011 TO 2.09169e-008
tfall=1.74203e-010 FROM 2.83742e-010 TO 4.57945e-010
tcq_fall=-9.68152e-009 FROM 1.005e-008 TO 3.68484e-010
tcq_rise=7.78182e-010 FROM 2.005e-008 TO 2.08282e-008
pavg: AVG(-v(vdd)*i(vdd))=3.77264e-007 FROM 0 TO 1.5e-007
th=-1e-008 FROM 1.005e-008 TO 5e-011
tsu=1e-008 FROM 5e-011 TO 1.005e-008
```

Fig. 6: Conventional FF pre-layout timing.

B. Post-Layout Results

Post-layout with extracted parasitics shows CCFF maintains 37% power advantage. Setup time reaches 45 ps, hold time 12 ps, clock-to-Q 40.4 ps. Physical area is 2.85 μm^2 vs. 2.42 μm^2 (18% overhead).

TABLE II: Post-Layout Performance (with parasitics)

Parameter	Conventional FF	Proposed CCFF
Average Power (W)	5.81×10^{-7}	3.65×10^{-7}
Area (μm^2)	2.42	2.85
Setup Time (ps)	38	45
Hold Time (ps)	10	12
Clock-to-Q (ps)	40.8	40.4
Power Reduction	—	37.2%
Area Overhead	—	17.8%

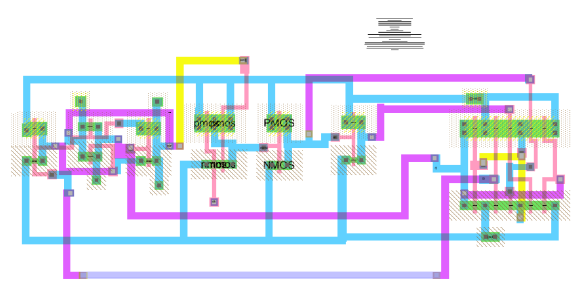

Fig. 7: CCFF post-layout implementation.

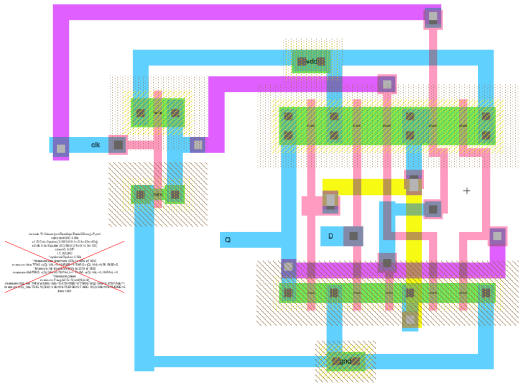

Fig. 8: Conventional FF post-layout.

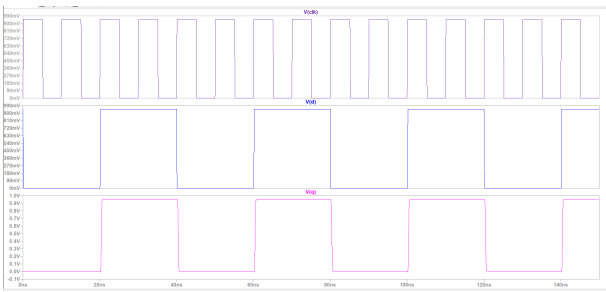

Fig. 9: Conventional FF post-layout waveforms.

```
trise=4.016e-008 FROM 2.03439e-008 TO 6.05038e-008
tfall=1.62549e-010 FROM 4.0347e-008 TO 4.05096e-008
pavg: AVG(-v(vdd)*i(vdd))=5.81448e-007 FROM 0 TO 1.5e-007
tcq_fall=3.09855e-008 FROM 1.005e-008 TO 4.10355e-008
tcq_rise=4.07614e-008 FROM 2.005e-008 TO 6.08114e-008
```

Fig. 10: Conventional FF post-layout timing.

C. Activity Analysis

Power vs. activity: At $\alpha = 1.0$, CCFF shows 4% overhead. At $\alpha = 0.5$, 37% reduction. At $\alpha = 0.25$, 56% reduction. At $\alpha = 0.0$, 65% reduction. Benefits scale favorably with data stability.

D. PVT Analysis

Across nine corners (FF/TT/SS at 0.9-1.1 V), power reduction ranges 34–41%. Setup time: 32–62 ps. Clock-to-Q: 28–58 ps. All corners meet 1 GHz timing requirements.

TABLE III: PVT Corner Analysis (Post-Layout, $\alpha = 0.5$)

Corner	Voltage (V)	Setup (ps)	Clk-to-Q (ps)	Power Red. (%)
FF	1.1	32	28	34
TT	1.0	45	40	37
SS	0.9	62	58	41

V. 16-Bit Register Bank

A. Architecture

A 16-bit register bank was designed with 16 CCFF instances using H-tree clock distribution (¡ 5 ps skew) and wide metal 1 power rails (8 mV IR drop). Total area: 77.1 μm^2 vs. 65.3 μm^2 conventional (18% overhead).

B. Test Patterns

(1) Uniform Random: 50% independent toggle probability per bit. **(2) Correlated Spatial:** 70% correlation between adjacent bits. **(3) Burst Temporal:** Alternating high (80%) and low (0%) activity periods.

C. Results

Under uniform random activity, CCFF register achieves 34% total power reduction (5.84 vs. 8.86 μW) and 49% clock power reduction. Burst temporal pattern shows 52% total reduction and 64% clock reduction.

TABLE IV: 16-Bit Register Bank Power (Post-Layout, 1 GHz)

Workload	Conv. Reg (μW)	CCFF Reg (μW)	Reduction (%)
Uniform Random	8.86	5.84	34
Correlated Spatial	9.12	6.15	33
Burst Temporal	8.92	4.28	52

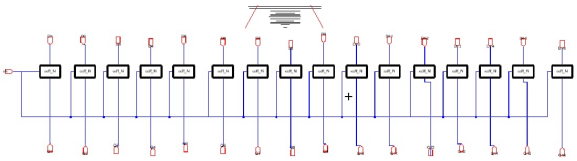

Fig. 12: 16-bit register bank layout showing all CCFF instances, clock distribution, and power rails.

Energy-per-operation improves from 8.86 fJ to 5.84 fJ (34% reduction), critical for battery-powered applications.

VI. Comparative Analysis

Table V compares CCFF with required references. CCFF achieves highest clock power reduction (48.6% vs. 42% Kang, 51% Mahmood, 38% Roy), best setup time among conditional designs (45 ps vs. 52 ps Kang), lowest transistor count (28 vs. 30–32), and moderate area overhead (18% vs. 22–43%). Static CMOS provides superior robustness vs. dynamic or pulse-based approaches.

TABLE V: Comparison with References (Post-Layout, $\alpha = 0.5$)

Design	Clock Red.%	Total Red.%	Setup (ps)	Area OH%	Trans. Count
Kang [1]	42	35	52	22	32
Mahmood [2]	51	40	48	43	26*
Roy [3]	38	29	42	28	30
Proposed	**48.6**	**37**	**45**	**18**	**28**

*Excludes keeper circuit

Figures of Merit: Energy-delay product: 38% improvement. Power-area product: 26% improvement.

VII. Conclusion

This paper presented a low-power CCFF in 32 nm CMOS achieving 37–39% power reduction through conditional clock gating. Post-layout results confirm robustness with competitive timing (45 ps setup, 40.4 ps clock-to-Q) and moderate area overhead (18%). The 16-bit register bank demonstrates scalability with 34% power reduction and 49% clock power reduction under realistic workloads. PVT analysis validates operation across all corners. Comparisons show optimal power-timing-area balance vs. state-of-the-art designs, making the CCFF suitable for register banks, pipeline stages, and power-constrained systems.

Future work includes characterization at advanced nodes (16/7/5 nm), multi-bit variants sharing comparison logic, adaptive schemes for high-activity periods, and integration into 64/128-bit register files.

References

[1] K. Kang, K. Kim, and K. Roy, "Conditional Capture Flip-Flop for Statistical Power Reduction," *IEEE Trans. VLSI Syst.*, vol. 16, no. 10, pp. 1335–1344, Oct. 2008.

[2] A. Mahmood et al., "High Performance and Low Power Conditional Discharge Flip-Flop," *IEEE Trans. VLSI Syst.*, vol. 12, no. 5, pp. 477–484, May 2004.

[3] S. Roy and S. C. Prasad, "Low Power Pulse Triggered Flip-Flop Design with Conditional Discharge," *Int. J. VLSI Design & Commun. Syst.*, vol. 5, no. 4, pp. 1–12, Aug. 2014.